

TEST - 13
GENERAL STUDIES - 3
ASCEND - GS MAINS TEST SERIES 2024

Q1: Give an account of rising Global Value Chains (GVCs) in India and examine the factors contributing to this growth. (10 Marks, 150 Words)

Approach to the answer:

Introduction: Briefly introduce Global Value Chains (GVCs) and India's participation.

Body: Discuss India's GVC growth trends, supported by key statistics, and analyze contributing factors like government initiatives, economic reforms, and global supply chain shifts.

Conclusion: Summarize India's GVC progress and future potential.

Answer: Global Value Chains (GVCs) refer to the international network of production, distribution, and marketing processes where different stages of goods and services are bought and sold across various countries. India's involvement in Global Value Chains (GVCs) rose from 10% in 1995 to 22% in 2019.

As of 2022, India's GVC participation stands at 40.3%, and India's GVC-related trade expanded nearly fourfold, from USD 62.9 billion in 2010 to USD 233.1 billion in 2022.

Factors Contributing to GVC Growth:

- **Government Initiatives:** Programs like Production Linked Incentive (PLI) and Districts as Export Hubs (DEH) initiative have directly incentivized higher value-added production and integration into global markets.
- **Economic Reforms:** Liberalization, simplification of regulatory frameworks, and the ease of doing business reforms have made India a more attractive destination for global firms.
- **Shifts in Global Supply Chains:** Geopolitical tensions and the desire for diversified supply chains have led firms to explore alternatives in Asia, with India emerging as a key beneficiary.
- **Technological Advancements:** Improvements in digital infrastructure and adoption of new technologies have enabled better integration of Indian firms into GVCs, particularly in high-value sectors.
- **Human Capital:** A growing, skilled workforce in areas like IT and manufacturing has supported India's increasing role in GVCs, especially in higher technology and service-oriented industries.

To sustain and further enhance its GVC participation, India needs to focus on improving trade infrastructure, integrating MSMEs into global networks, simplifying entry and exit procedures, and implementing effective trade facilitation measures. These steps will ensure that India remains competitive and continues to grow its presence in global value chains.

Q2: Examine the necessity of a well-developed care economy in India and highlight its potential to unlock a new economic segment. (10 Marks, 150 Words)

Approach to the answer:

Introduction: Briefly define the care economy and its components.

Body: Discuss the necessity of a robust care economy in India and how it unlocks a new economic segment.

Conclusion: Summarize by benefits of a developed care economy or on a positive note.

Answer: Care work encompasses activities aimed at addressing the physical, psychological, and emotional needs of individuals across various age groups and conditions, from children to the elderly.

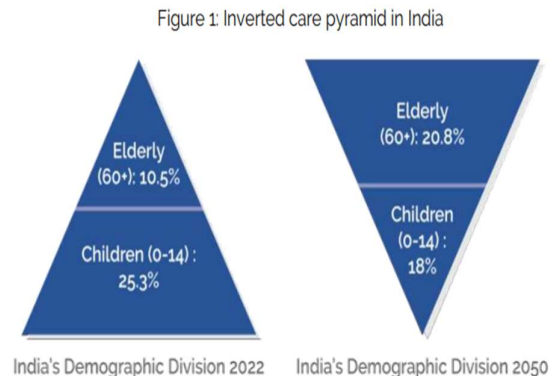
Care work can be categorized into two types:

- **Unpaid or Underpaid Work:** This includes caregiving and social support tasks often carried out within households, such as childcare, eldercare, and domestic work.

- **Paid Care Work:** This involves labor performed for remuneration by professionals such as nurses and caregivers.

Increasing Need for a Well-Developed Care Economy

- **Demographic Transition:** By 2050, the total population needing care will increase from 50.7 crore in 2022 to 64.7 crore in 2050.
- **Equal Opportunities for Women:** Developing a care economy can help decouple gender roles from unpaid care work, thereby improving female labor force participation rates (FLFPR).
 - Women disproportionately bear the burden of unpaid care work, contributing to lower FLFPR globally, including in India.
 - On average, working-aged women in India spend 5.6 hours per day on unpaid work, compared to 30 minutes for men.
 - Women in paid employment also face a 'double burden,' spending nearly six times as much time on unpaid care work compared to men.
 - The 'motherhood penalty' refers to the significant career costs women face due to childbearing and childcare responsibilities.
- **Economic Potential:** Investing in the care services sector could generate approximately 475 million jobs globally by 2030. In India, a direct public investment equivalent to 2% of GDP has the potential to create 11 million jobs, with nearly 70% of these opportunities going to women.



Care economy unlock a new economic segment in India:

- **Job Creation:** Investment in care services creates new jobs, especially for women, driving economic growth.
- **Boosting Consumer Demand:** Expanding care services increases demand for related products, stimulating local economies.
- **Formalizing Informal Work:** Shifting unpaid care work to paid services formalizes the sector, expanding the tax base.
- **Attracting Investment:** Development of care infrastructure invites public and private investments, fostering new business opportunities.
- **Expanding paid Care:** Expanding paid care services reduces women's unpaid care burden, increasing their workforce participation.
- **Boosting Female Employment:** Formalizing care work creates paid jobs for women, enhancing economic independence.

A well-developed care economy can unlock a new economic segment by addressing critical societal needs, promoting gender equality, driving job creation and economic growth.

Additional Information:

Note: The silver economy includes all those economic activities, products and services designed to meet the needs of the elderly population. According to the 'India Ageing Report 2023' of the United Nations Population Fund (UNFPA), there were 149 million persons (10.5%) aged 60 years and above in 2022 (as of | July). And, by 2050, this population will double to 20.8%, with the absolute number at 347 million.

Measures to strengthen Care Economy in India:

- **Financial Support for Maternity Leave:** Introduce financial assistance for MSMEs and Start-Ups to cover the costs of six months paid maternity leave, following practices in countries like Japan and France.
- **Enhancing Support for Care Services:** Expand financial support for care services under Mission Shakti,

targeting SHGs for childcare and eldercare, and provide tax incentives for women-led MSMEs offering care services.

- **Boosting Care Infrastructure Investment:** Increase public and private sector investment in care facilities, especially in rural areas. Expand on existing schemes like Saksham Anganwadi and POSHAN 2.0 to improve infrastructure.
- **Developing Care Worker Skills:** Focus on skilling, reskilling, and upskilling care workers to address the supply-demand gap and enhance the quality of care. Address challenges like low wages and high informality in the sector.
- **Quality Assurance Systems:** Establish formal regulatory mechanisms for maintaining care service standards in India, similar to the OFSTED system in the UK, to ensure high-quality care services.

Q3: Discuss the significance of the Agro-processing sector in promoting rural employment and economic growth in India. (10 Marks, 150 Words)

Approach to the answer:

Introduction: Briefly introduce the agro-processing sector.

Body: Discuss how agro-processing enhances rural employment and economic growth by adding value to agricultural products. Also, suggest measures to support sector growth.

Conclusion: Summarize the sector's potential to drive rural economic development and job creation.

Answer: The agro-processing sector involves the transformation of raw agricultural products into processed goods for consumption or further manufacturing. The agro-processing sector presents opportunities for rural development by facilitating the transition from farm to factory, thus creating ample job prospects in India.

Significance of Agro-Processing

- **Rural Job Creation:** Agro-processing can enhance productivity, support crop diversification, and provide extensive employment opportunities, fostering rural economic growth.
- **Low Value Addition:** India's food processing levels are notably low—4.5% for fruits, 2.7% for vegetables, 21.1% for milk, 34.2% for meat, and 15.4% for fishery—compared to 30% in China, highlighting the need for improvement.
- **Growing Market Demand:** The Indian food processing market is projected to reach USD 535 billion by 2025, with a CAGR of 15.2%, driven by increasing demand for diverse and local food products.
- **Successful Models:** Notable examples such as the Sahyadri Farmer Producer Company in Maharashtra and Araku Coffee plantations in Andhra Pradesh demonstrate the potential of agro-processing in rural development.
- **Captive Demand Opportunities:** Local agro-processing units can supply mid-day meals in schools and contribute to health consciousness and local pride through local agglomerations.
- **Program Synergies:** The sector can benefit from integrating existing programs like Mega Food Park, Skill India, Mudra, and the One District-One Product scheme for better labor, logistics, credit, and marketing support.

Measures to support the growth of the agro-processing sector:

- **Infrastructure Development:** Improve facilities like cold storage, transportation networks, and processing units to enhance efficiency and reduce post-harvest losses.
- **Financial Support:** Provide access to credit and subsidies for small and medium-sized enterprises (SMEs) and farmers involved in agro-processing.
- **Policy Framework:** Develop supportive policies, such as tax incentives and streamlined regulations, to encourage investment and ease the operational challenges faced by agro-processing units.
- **Market Linkages:** Facilitate better access to domestic and international markets through marketing support, branding initiatives, and trade fairs.

Notable schemes such as **Pradhan Mantri Kisan SAMPADA Yojana (PMKSY)**, **Production Linked Incentive Scheme for Food Processing Industry (PLISFPI)**, etc. are steps in right direction to realize the potential of agro-processing sector in India.

Q4: Analyze the role of Direct Benefit Transfer (DBT) in enhancing the resilience among Indian farmers. Also, identify key issues associated with it. (10 Marks, 150 Words)

Approach to the answer:

Introduction: Introduce Direct Benefit Transfer (DBT) and its purpose.

Body: Analyze how DBT improves financial stability and resilience for Indian farmers. Suggest measures to optimize DBT's impact.

Conclusion: Summarize DBT's role in enhancing farmers' resilience.

Answer: Direct Benefit Transfer (DBT) is a government mechanism for transferring subsidies and benefits directly to eligible beneficiaries' bank accounts. Key agricultural schemes under DBT include the PM-Kisan Scheme, Pradhan Mantri Fasal Bima Yojana, etc

DBT enhances resilience among Indian farmers by providing:

- **Financial Stability:** Regular income transfers offer farmers a stable and predictable revenue stream.
 - Example: PM-Kisan Yojana.
- **Targeted Support:** Direct transfers ensure that assistance reaches the most vulnerable farmers in need.
- **Flexibility in Resource Allocation:** Farmers can allocate funds based on their specific needs, such as purchasing better seeds or improving irrigation.
- **Timely Assistance and Insurance:** DBT facilitates quick financial relief during crises like droughts or floods, helping farmers recover and continue their activities.
 - Example: PM Fasal Bima Yojana.

Current issues with the Direct Benefit Transfer (DBT) Scheme:

- **Exclusion Errors:** Problems with identification or incorrect data can lead to exclusion errors.
 - Example: Farmers may struggle to receive benefits if their Aadhaar details are not correctly linked.
- **Insufficient Assistance:** For example, Rs. 6,000 annual assistances through PM KISAN SAMMAN are deemed inadequate, covering only 9.65% of total crop costs.
- **Inadequate Coverage of Tenant Farmers:** Such DBT schemes mainly benefit land-owning farmers, often excluding tenant farmers who do not own land.
- **Land Ownership Documentation:** Farmers lacking proper land ownership documents, such as tenant farmers or those on leased land, may face difficulties accessing benefits.
- **Tech-Driven Challenges:** Low digital literacy among farmers can hinder their ability to access scheme benefits.

DBT enhances resilience among Indian farmers by ensuring timely and efficient financial support, reducing corruption, facilitating access to subsidies, and fostering financial inclusion. This approach strengthens farmers' ability to manage risks and invest in their agricultural activities.

Q5: Discuss the key features of Pradhan Mantri Kaushal Vikas Yojana 4.0. How do these features contribute to addressing the skill gap in India's workforce? (10 Marks, 150 Words)

Approach to the answer:

Introduction: Briefly introduce PMKVY 4.0.

Body: Discuss the key features of PMKVY 4.0, highlighting its focus areas. Explain how these components help bridge India's skill gap.

Conclusion: Summarize how PMKVY 4.0 addresses workforce skill needs.

Answer: PMKVY 4.0 is an advanced skill development initiative under the 'Skill India Programme'. Building on the outcomes of its previous phases, the program aims to establish a robust skill ecosystem to address current challenges and the evolving needs of industries.

Features of PM Kaushal Vikas Yojana 4.0

- **Enhanced Industry-Connect:** Greater reliance on On-the-Job Training for hands-on learning.
 - Encourages industry participation and contribution to skill development.
- **Emphasis on Re-skilling and Upskilling:** Focus on Recognition of Prior Learning (RPL) with innovative assessment and monitoring.
- **Digitalization of the Training Ecosystem:** Integration of online systems for registration and counselling.
- **Simplification of Procedures:** For reduction of compliance burden.
 - Enlistment of multiple Assessment Agencies and Awarding Bodies.
- **Creation of a National Pool of Trainers:** Ensures availability and efficient implementation of training.
 - Facilitates faster assessment.
- **Focus on New Age Skills:** Training in emerging fields like Industry 4.0, Web 3.0, AI/ML, AR/VR, Climate Change, Circular Economy, Green Economy, and Energy Transition.
- **Focus on Informal Sector and Self-Employment:** Training job roles addressing skill needs of the informal sector and self-employed.

PM Kaushal Vikas Yojana 4.0 address the skill gap in India's workforce in the following ways:

- **Enhanced Industry-Connect:** By increasing on-the-job training and industry participation, the program aligns skills with industry needs, ensuring that the workforce is equipped with relevant, practical skills.
- **Emphasis on Re-skilling and Upskilling:** Focusing on Recognition of Prior Learning (RPL) acknowledges existing skills and promotes continuous learning, enabling workers to adapt to evolving job requirements.
- **Digitalization of the Training Ecosystem:** Integrating online systems for registration and counselling makes training more accessible and efficient, broadening participation.
- **Simplification of Procedures:** Reducing compliance burdens and enlisting multiple assessment agencies streamline the certification process, enhancing skill recognition and mobility.
- **Creation of a National Pool of Trainers:** This ensures consistent and quality training delivery, addressing regional disparities in skill development.
- **Focus on New Age Skills:** Training in emerging technologies and sectors prepares the workforce for future job markets, reducing the skills gap in high-demand areas.
- **Focus on Informal Sector and Self-Employment:** By addressing the skill needs of the informal sector and self-employed individuals, the program ensures inclusivity, helping a broader section of the workforce to enhance their skills and employability.

PM Kaushal Vikas Yojana 4.0 is a key initiative in India's skill development landscape, focusing on industry alignment, technological integration, and inclusivity to close the skilled labour gap. It aims to effectively leverage India's demographic dividend.

Q6: Explain the working principle of 3D printing technology. Also, discuss its applications across various sectors. (10 Marks, 150 Words)

Approach to the answer:

Introduction: Briefly introduce 3D printing technology and its working principle.

Body: Explain the step-by-step process of how 3D printing works. Discuss its applications in sectors like healthcare, automotive, aerospace, and education.

Conclusion: Summarize the impact and potential of 3D printing technology across various fields.

Answer: 3D printing, also known as additive manufacturing, is a process of creating three-dimensional objects by layering materials such as plastic, metal, or ceramics, based on a digital design.

Working Principle of 3D Printing Technology: 3D printing technology utilizes computer-aided design (CAD) software to create a digital model of the object to be printed.

- This model is sliced into thin layers by the software, which then sends instructions to a 3D printer.
- The printer deposits material layer by layer, following these instructions, until the object is fully formed. Various materials, including plastics, metals, ceramics, and even living cells, can be used in the process.

Steps in 3D Printing:

- **Creation of a 3D Design:** The process begins with creating a 3D design using CAD software. This design is a digital blueprint specifying the object's dimensions and geometry.
- **Preparation of the 3D Model:** The digital design is converted into a format compatible with 3D printing software. The 3D model is then sliced into hundreds or thousands of horizontal layers, each representing a cross-section of the final object.
- **Selection of Materials:** The choice of material depends on the final object's requirements. Common materials include plastic, metal, ceramics, and biomaterials.
- **3D Printing Process:** The printer builds the object layer by layer, depositing material precisely according to the design specifications.
- **Post-Processing:** Once the object is printed, it may require post-processing to remove support structures, smooth surfaces, or add finishing touches, such as sanding, polishing, or painting.

Potential Applications of 3D Printing Technology in Various Sectors of the Indian Economy:

- **Healthcare:** 3D printing enables the creation of customized prosthetics, implants, and even organs, which can reduce mortality rates and make treatments more affordable.
 - For Example: Creation of personalized orthopedic implants tailored to a patient's unique anatomy.
- **Architecture and Construction:** 3D printing can develop complex architectural models and even entire buildings, making construction activities more sustainable and time efficient.
 - For Example: Using 3D printing to create affordable and sustainable housing for rural communities.
- **Education and Research:** Educational institutions can use 3D printing to create anatomical models for medical education and complex scientific models, enhancing the learning process.
 - For Example: Students designing and printing prototypes of prosthetic limbs or wind turbine blades.
- **Aerospace and Defense:** In aerospace and defense, 3D printing can create lightweight and complex parts, boosting security and space technology.
 - For Example: The Indian Space Research Organization (ISRO) has used 3D printing to create rocket engine components.
- **Automotive Industry:** The automotive sector can use 3D printing to fabricate customized and lightweight components, improving efficiency, reducing weight, and lowering costs.
 - For Example: Mahindra & Mahindra using 3D printing to create automotive parts and prototypes.

3D printing technology has the potential to revolutionize various sectors of the Indian economy. The government has introduced the "National Strategy on Additive Manufacturing" to harness the potential of 3D

printing technology, aiming to make India a manufacturing hub, attract foreign direct investment (FDI), generate employment opportunities, and achieve self-sufficiency in these sectors.

Q7: Discuss the impact of rapid urbanization on the deterioration of water quality of the wetlands. What steps can be taken to address the issue of eutrophication in urban wetlands? (10 Marks, 150 Words)

Approach to the answer:

Introduction: Briefly introduce the impact of rapid urbanization on wetland water quality.

Body: Discuss how urbanization contributes to water deterioration and eutrophication. Suggest steps to mitigate eutrophication in urban wetlands.

Conclusion: Summarize measures to protect wetland ecosystems.

Answer: According to Wetlands International South Asia, nearly 30 percent of India's natural wetlands have been lost in the past three decades due to illegal construction, unsustainable urbanization, and agricultural expansion. Additionally, the increased occurrence of eutrophication has worsened water quality, as seen in cases like the lake adjacent to the Perungudi dumpyard in Chennai turning pink.

Rapid urbanization adversely affects wetland water quality in several ways:

- **Industrial Pollution:** The establishment and expansion of industries near wetlands release pollutants such as heavy metals, chemicals, and untreated effluents into nearby water bodies.
 - For Example: Kasadi River near Mumbai.
- **Domestic Wastewater:** Growing urban populations produce substantial amounts of domestic wastewater, which is often discharged into wetlands without proper treatment.
 - For Example: Bellandur Lake in Bangalore, known for urbanization-induced water pollution.
- **Surface Runoff:** Urban expansion creates impervious surfaces that increase surface runoff, carrying pollutants like sediments, heavy metals, and nutrients into wetlands.
 - For Example: Adyar River in Chennai, affected by pollutants from streets and stormwater drains.
- **Nutrient Runoff:** In peri-urban areas where agriculture and urban development intersect, fertilizers and pesticides contribute to nutrient runoff into wetlands.
 - For Example: Wetlands around Hesaraghatta Lake in Bangalore.
- **Habitat Loss:** Urban expansion often encroaches on wetland areas for real estate development, resulting in habitat loss and disruption of natural ecosystems.
 - For Example: Pallikaranai Marsh in Chennai.
- **Solid Waste Management:** Inadequate solid waste management practices lead to the disposal of solid waste into wetlands, causing water pollution.

Rapid urban development introduces untreated sewage and industrial effluents into urban wetlands, leading to severe frothing and toxic blooms, a phenomenon known as eutrophication (overabundance of nutrients in a water body). This significantly impacts water quality and the surrounding ecosystem.

Measures to Tackle Eutrophication in Urban Wetlands:

- **Upgrade Wastewater Treatment:** Enhance existing wastewater treatment plants with advanced technologies to effectively remove nutrients. For Example: Upgraded Okhla Sewage Treatment Plant.
- **Phytoremediation Techniques:** Use aquatic plants to absorb excess nutrients from water. For Example: Water hyacinth used in Dal Lake, Srinagar.
- **Constructed Wetlands:** Implement constructed wetlands for tertiary treatment of wastewater before it is released into urban wetlands. For Example: Eco-Pond in Kolkata.
- **Green Infrastructure:** Incorporate permeable surfaces and green spaces in urban planning to reduce stormwater runoff and nutrient transport. For Example: Sabarmati Riverfront Development Project in Ahmedabad.
- **Riparian Vegetation:** Restore and protect vegetation along wetland edges to act as a buffer against nutrient runoff. For Example: Restoration of Adyar Creek in Chennai.

- **Community Involvement:** Conduct awareness campaigns and engage local communities in monitoring water quality and reporting pollution incidents, fostering a sense of responsibility. For Example: Loktak Lake in Manipur.
- **Wastewater Recycling:** Recycle and reuse treated wastewater for irrigation to prevent further contamination of groundwater. For Example: Seechewal model in Punjab.

Such an integrated strategy can mitigate the impacts of rapid urbanization on wetland ecosystems and promote their long-term health and sustainability.

Q8: Highlight factors contributing to the increasing frequency and intensity of heat waves in India. Also, suggest measures to improve resilience and reduce the vulnerability of the population at risk.

(10 Marks, 150 Words)

Approach to the answer:

Introduction: Briefly introduce the rising frequency and intensity of heat waves in India or with definition of Heat wave.

Body: Highlight contributing factors and suggest measures to enhance resilience and reduce vulnerability of at-risk populations.

Conclusion: Summarize key strategies for adaptation and mitigation.

Answer: Heat waves are periods of unusually high temperatures exceeding the normal maximum during the summer season. Heatwaves are recognized when the maximum temperature of a station reaches at least 40°C for Plains and 30°C for Hilly regions.

Heat waves in India are becoming more frequent and intense due to climate change, significantly impacting the health, livelihood, and overall well-being of its population.

Key factors contributing to Heat Wave:

- **Rising Global Temperatures:** Increased greenhouse gas emissions have led to a rise in global temperatures, intensifying heat waves.
- **Urban Heat Island Effect:** Rapid urbanization has led to reduced green cover and increased heat absorption by concrete structures, amplifying the urban heat island effect.
- **Changes in Atmospheric Circulation:** Altered weather patterns due to climate change, such as delayed monsoons and longer dry spells, contribute to prolonged heatwave conditions.
- **Anticyclonic Flow:** Anti-cyclones are large-scale atmospheric circulation patterns characterized by sinking air. This **sinking air compresses and warms adiabatically** (heats up without gaining heat), contributing to rising temperatures at the surface.
- **Absence of Moisture:** Moisture in the air retains heat and prevents it from escaping while dry air allows solar radiation to reach the ground more readily and radiate back out at night with minimal trapping, resulting in a quicker increase in daytime temperatures.
- **Water Scarcity and Drought:** Reduced water availability exacerbates heat stress, particularly in rural areas that rely on agriculture.
- **Geographical Considerations:** Heat waves are more frequent in regions with arid or semi-arid climates, such as Northwest India. Prevailing westerly winds during summer months can spread heat waves eastward and southward.

To enhance resilience against heat waves in India, the following strategies can be implemented:

- **Urban Planning and Infrastructure:** Develop heat-resilient infrastructure, like green roofs and urban forestry, to reduce temperatures and mitigate the urban heat island effect.
- **Heat Action Plans (HAPs):** Expand and refine HAPs to include community-specific adaptations, early warning systems, and targeted measures for vulnerable populations.
- **Community Awareness and Engagement:** Conduct community training and outreach programs to raise awareness about heat risks and safety measures.

- **Capacity Building in Healthcare:** Train healthcare workers to handle heat-related illnesses and ensure facilities are prepared for patient surges during heat waves.
- **Funding and Resource Allocation:** Increase funding for heat resilience initiatives and prioritize heat-related policies through government budgets and public-private partnerships.
- **Multi-Sector Coordination:** Foster coordination among government departments, NGOs, and community organizations to streamline heat wave management efforts.

By integrating these strategies and notifying heat waves as Natural Disaster to bring it under legal framework of the Disaster Management Act 2005 to safeguard vulnerable populations against the increasing frequency and intensity of heat waves.

Additional Information:

NDMA Guidelines for Heatwaves:

- Avoid going out in the sun, especially between 12.00 noon and 3.00 p.m.
- Drink sufficient water as often as possible.
- Wear lightweight, light-coloured, loose, and porous cotton clothes. Use protective goggles, an umbrella/hat, shoes or chappals while going out in sun.
- Avoid alcohol, tea, coffee and carbonated soft drinks, which dehydrate the body.
- Avoid high-protein food and do not eat stale food.
- Use ORS, homemade drinks like lassi, torani (rice water), lemon water, buttermilk, etc. which helps to rehydrate the body.
- Keep animals in the shade and give them plenty of water to drink.
- Keep your home cool, use curtains, shutters or sunshade and open windows at night.
- Use fans, damp clothing and take baths in cold water frequently.

Q9: 'Damage to critical infrastructure greatly intensifies the effects of natural disasters.' In this context, discuss the importance of constructing disaster-resilient infrastructure in India. (10 marks, 150 words)

Approach to the answer:

Introduction: Briefly introduce the concept of disaster-resilient infrastructure.

Body: Discuss how damage to critical infrastructure exacerbates natural disasters. Highlight the importance of resilient infrastructure in minimizing losses and ensuring swift recovery.

Conclusion: Summarize the need for resilient infrastructure in disaster-prone India.

Answer: Natural disasters, such as heat waves, floods, cyclones, droughts, and cloudbursts, are becoming more frequent and severe due to climate change.

When disasters impact critical infrastructures like power, telecommunications, transportation, health services, and cyber systems, they can severely hinder relief, rescue, and recovery efforts. For example, an earthquake can cause extensive damage to medical facilities, affecting public health and complicating emergency response efforts. Therefore, damage to critical infrastructure can significantly amplify the effects of natural disasters.

Significance of Disaster-Resilient Infrastructure

- **Protection of Human Lives:** Incorporating disaster-resilient infrastructure is crucial for saving lives. For instance, earthquake-resistant buildings and bridges can prevent collapses and reduce casualties.
- **Economic Impact:** The increasing frequency and intensity of disasters have led to escalating economic losses. Between 2018 and 2023, states in India spent over ₹1.5 lakh crore on post-disaster recovery and rehabilitation, according to government reports.
- **Cost Reduction:** Resilient infrastructure minimizes the need for expensive repairs and ensures the continuity of economic activities, reducing the financial burden of disasters.

- **Community Resilience:** Disaster-resilient infrastructure enables communities to withstand and recover from disasters more effectively, enhancing social stability and reducing the vulnerability of rural and economically weaker regions.
- **Livelihood Protection:** Natural disasters significantly impact livelihoods. A 2022 World Bank report estimated that heat-related stress could reduce productivity and potentially result in the loss of around 34 million jobs in India by 2030.
- **Climate Change Adaptation:** Building disaster-resilient infrastructure is a crucial adaptation strategy to mitigate the adverse effects of climate change, protecting communities and economies from its impacts.
- **Alignment with Global Goals:** Disaster-resilient infrastructure supports the Sustainable Development Goals (SDGs), particularly SDG 9 (Industry, Innovation, and Infrastructure) and SDG 11 (Sustainable Cities and Communities). It also aligns with the Sendai Framework for Disaster Risk Reduction and the 2030 Agenda.

Strategies for Building Disaster-Resilient Infrastructure

- **Framework Development:** Develop frameworks and governance mechanisms at national and sub-national levels for constructing disaster-resilient infrastructure based on local susceptibility to disasters. For example, Odisha, prone to cyclones, has fragile infrastructure, with many power stations and electricity poles vulnerable to high wind speeds, as highlighted by the CDRI report.
- **Disaster Management Plans:** Infrastructure sectors should implement disaster management plans to enhance preparedness and response. For instance, hospitals in disaster-prone areas should have backup power supplies, airports and railways should have measures to prevent or quickly drain waterlogging, and telecommunication lines should be buried underground.
- **Regular Inspections and Retrofitting:** Conduct regular structural health inspections, seismic retrofitting of buildings in earthquake-prone regions, and integrate early warning systems and preparedness plans.
- **Urban Planning and Policy:** Incorporate disaster-resilient infrastructure in urban planning and policymaking by enforcing building codes, land-use planning, and investing in research and innovation for resilient infrastructure technologies.
- **Collaboration:** Foster collaboration among government, the private sector, and civil society to ensure comprehensive and inclusive disaster risk management.

Building disaster-resilient infrastructure is a strategic necessity for India, given the increasing frequency and intensity of natural disasters. It is vital for safeguarding lives, reducing economic losses, ensuring the continuity of essential services, and supporting sustainable development.

Q10: India has vast coastline and with growing centrality of Indian Ocean region, it has become pertinent to remove inbuilt deficiencies in our coastal security approach. Elaborate. (10 Marks, 150 Words)

Approach to the answer:

Introduction: Briefly introduce India's vast coastline and its strategic importance in the Indian Ocean region.

Body: Discuss the deficiencies in India's coastal security. Elaborate on the need for enhanced measures and reforms to strengthen coastal security.

Conclusion: Summarize the importance of a robust coastal security framework.

Answer: India has a vast coastline of 7500 Kms making it vulnerable to sea-borne traditional and non-traditional security threats. At the same time, Indian Ocean Region has become the focal point of global geopolitics, complicating India's security scenario.

Growing centrality of Indian Ocean Region:

- The region lies along key Sea Lanes of Communication with 75 percent of the world's maritime trade and 50 percent of daily global oil transfers passing through the region.
- It has Choke Points like Strait of Malacca, Strait of Hormuz, and Bab al-Mandab.
- Emergence of Indo-Pacific as new geographical construct.

- Chinese adventurism in the region and infrastructure build-up under Belt and Road Initiative.
- Emerging traditional and non- traditional threats like terrorism, piracy, human and drug trafficking etc.

Inbuilt deficiencies in our coastal security approach:

- Multiple agencies: - lack of accountability and coordination; Lack of Apex Maritime Authority
- Two of the recent CAG reports reveal a number of shortcomings like Delays in creation of shore-based infrastructure, under-utilization of petrol boats, Manpower shortage, resource deficit, unspent funds etc.
- Lack of a clear maritime policy
- IB audit in 2016: out of 227 minor ports in India 187 had little or no security at all.
- Infrastructure constrains and no effective mechanism to monitor smaller boats. most of them are without any transponders.

Steps to be taken for robust coastal security:

- Increased Surveillance and inter-agency coordination: -ensure mandatory fitment of AIS on power-driven vessels with a length of more than 10 m.
- Stronger involvement of Coastal Police and Indian coast Guards: -strengthening and better integrating coastal police and guards into littoral security architecture.
- Develop a robust infrastructure and tracking system for all the fishing vessels: - vessels should be fitted with transponders
- 5 Point Agenda on Maritime Security: The 5-point agenda on maritime security by the UNSC should be implemented in letter and spirit.

Steps Taken:

- Under Coastal Security Scheme, coastal states/UTs have operationalized 127 coastal police stations.
- 10 coastal police stations in various coastal states/UTs have been notified to deal with crimes committed in international waters.
- 1382 offshore islands have been identified for their holistic development.
- National Academy of coastal policing (NACP) is being setup at Mojar village, dist- Devbhoomi in Gujarat.
- 500 satellite-based transponders are being installed on fishing boats below 20 metres of length in sensitive areas of Gujarat and Tamil Nadu for location tracking.

Supplementing these initiatives, there is need to leverage technologies like AI, IOT along with proper training, evaluation, monitoring to fill the gap and ensure robust coastal security framework that will reflect India is a responsible maritime power in the region.

Q11: Critically assess the achievements and challenges associated with the implementation of the Fiscal Responsibility and Budget Management (FRBM) framework. (15 Marks, 250 Words)

Approach to the answer:

Introduction: Briefly introduce the FRBM framework.

Body: Assess the achievements of the FRBM framework and the challenges faced in its implementation.

Conclusion: Summarize the overall impact of the FRBM framework or conclude with the way forward.

Answer: Fiscal discipline involves maintaining a balanced and sustainable government budget by controlling spending and limiting borrowing, which ensures smooth monetary operations and macroeconomic stability.

Ways in which fiscal discipline contributes to overall macroeconomic stability:

- **Prudent Financial Management:** Avoiding excessive borrowing and ensuring responsible taxation policies allow governments to build fiscal reserves for crises, such as the COVID-19 pandemic.
- **Responsible Debt Management:** Prioritizing spending, avoiding deficit financing, transparent borrowing, and structural reforms to improve revenue collection aid in managing debt effectively.

- **Sustainable Economic Growth:** Consistent economic policies and efficient resource allocation build market confidence and sustain growth. Investments in infrastructure and public-private partnerships (PPPs) drive GDP growth, e.g., increased capital expenditure on infrastructure.

Achievements of the FRBM Framework:

- **Deficit Reduction and Debt Control:** The FRBM framework successfully reduced the fiscal deficit and total liabilities, e.g., India lowered its fiscal deficit from 4.5% of GDP in 2003-04 to 3.5% in 2019-20 and reduced the debt-to-GDP ratio from 75.9% to 68.7% in the same period.
- **Structural Reforms:** The framework led to significant reforms focusing on sustainable growth over short-term measures, e.g., simplification of the income tax structure and implementation of GST.
- **Transparent Fiscal Policy:** FRBM mandates increased transparency in fiscal operations, e.g., the "Medium Term Fiscal Policy Statement" presented with the Budget.
- **Impact on Economic Indicators:** Reduced government borrowing has boosted the private sector, leading to job creation, lower interest rates, and improved GDP, with better fiscal indicators over time.
- **Cyclical Flexibility:** Amendments to the FRBM allowed for adjustments during downturns, e.g., the escape clause invoked during the COVID-19 pandemic.

Challenges Associated with Implementation of the FRBM Act:

- **Public Investment Constraints:** Strict rules may limit public investment, impacting critical infrastructure and long-term growth, e.g., social programs for education, health, and poverty alleviation.
- **Economic Downturns:** Adhering to deficit targets during downturns restricts fiscal stimulus, worsening recessions, e.g., during the COVID-19 crisis.
- **Global Economic Conditions:** External shocks can affect adherence to FRBM targets, requiring flexibility, e.g., disruptions from the Russia-Ukraine conflict.
- **Off-Budget Financing:** Use of off-budget financing to meet targets undermines transparency and effectiveness.
- **Inadequate Monitoring:** Inconsistent monitoring across states affects the framework's effectiveness.
- **Political Pressures:** Short-term political considerations can lead to deviations from FRBM targets, especially during elections, e.g., subsidies and loan waivers.
- **Poor State Finances:** High revenue deficits and debt ratios in some states counteract the central government's FRBM measures, e.g., West Bengal, Haryana, Punjab.

Way Forward to Strengthen the FRBM Act:

- **Flexible Fiscal Rules:** Allow for flexibility in targets to implement counter-cyclical measures during downturns.
- **Medium-Term Expenditure Framework (MTEF):** Project expenditures and revenues over the medium term for better planning.
- **Enhanced Transparency:** Improve reporting on off-budget financing, contingent liabilities, and fiscal risks. Regularly review the FRBM framework for relevance.
- **Independent Fiscal Council:** Establish an unbiased council to assess fiscal policies.
- **Incentivize States:** Reward states demonstrating fiscal discipline and share best practices to promote uniformity.

Following the recommendations by the NK Singh FRBM review committee will make the FRBM framework more adaptable to changing economic conditions.

Q12: Give an account of capital expenditure on infrastructure projects in India. Also, highlight the key challenges in mobilizing private investment, and how can Public-Private Partnerships (PPPs) address these challenges effectively?
(15 Marks, 250 Words)

Approach to the answer:

Introduction: Briefly introduce the concept of capital expenditure on infrastructure projects in India.

Body: Discuss the current state of capital expenditure and key challenges in mobilizing private investment. Highlight how PPPs can address these challenges through risk-sharing and enhanced efficiency.

Conclusion: Summarize the role of PPPs in overcoming investment challenges and fostering infrastructure growth.

Answer: Capital expenditure by the government refers to spending on long-term investments like infrastructure, assets, or projects that enhance future economic capacity, often financed through borrowing or budget allocations.

Data:

- The Union Government's capital expenditure on infrastructure has increased 2.2 times from FY21 to FY24.
- State governments also saw a 2.1-fold rise in capital expenditure from FY21 to FY24.
- The share of gross budgetary support to key sectors like Railways and the National Highway Authority of India (NHAI) increased from 36.1% in FY21 to 42.9% in FY24.
- For FY25, the central government has allocated Rs 11.11 lakh crore (3.4% of GDP) for capital expenditure, reflecting a 31.6% increase in support for state governments and institutions.

Key Challenges in Mobilizing Private Investment:

- **Low Bank Credit:** Bank credit for infrastructure was only Rs 79,000 crore from March 2023 to March 2024, much less than the government's allocation.
- **Sector Concentration:** Bank credit is concentrated in a few sectors like roads, airports, and power.
- **Limited Debt and Equity Issuances:** In FY24, capital market issuances for infrastructure were only Rs 1 lakh crore.
- **Regulatory and Risk Uncertainty:** Evolving regulatory frameworks and risk-sharing issues deter private investment.
- **Funding Shortages:** External Commercial Borrowing (ECB) and Real Estate Investment Trusts (REITs) show limited inflow compared to government budgets.

Role of Public-Private Partnerships (PPPs):

- **Diversifying Funding Sources:** PPPs can mobilize additional funds by attracting private investment.
- **Expertise and Efficiency:** Private sector involvement brings innovation and efficiency to infrastructure projects.
- **Risk Sharing:** PPPs distribute risks between the government and private entities, making projects more feasible.
- **Enhanced Project Viability:** Initiatives like the Public Private Partnership Appraisal Committee (PPPAC) and the India Infrastructure Project Development Fund (IIPDF) support project development and viability, enhancing PPP effectiveness.
- **Reducing Reliance on Public Expenditure:** By leveraging private investments, PPPs reduce the sole dependence on government funding for infrastructure growth.

India should continue to harness the potential of PPPs, addressing regulatory uncertainties and improving risk management. This strategy will help sustain infrastructure development and balance public and private funding sources.

Q13: Analyze the evolution of India's financial inclusion strategy, highlighting its key achievements and challenges. Also, suggest measures to improve its effectiveness. (15 Marks, 250 Words)

Approach to the answer:

Introduction: Briefly introduce the role of digital financial tools in enhancing financial inclusion.

Body: Discuss India's evolving approach to digital financial inclusion, highlighting key achievements and challenges.

Conclusion: Summarize India's progress by leveraging digital tools for financial inclusion.

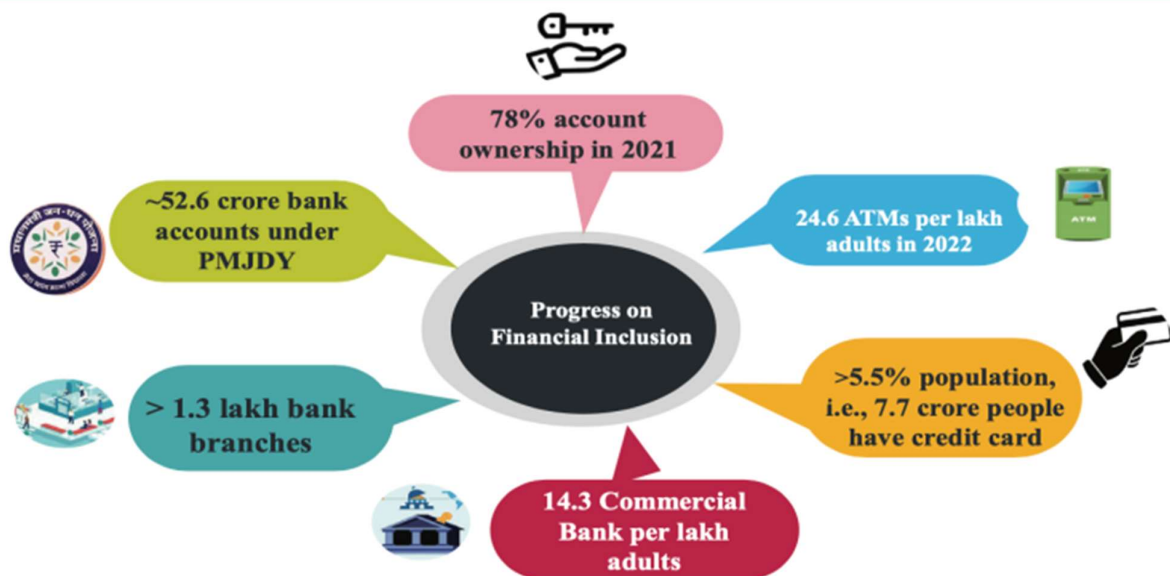
Answer: Financial inclusion is ensuring access to affordable financial services for all individuals, especially the underserved, to promote economic participation and stability. Digital financial tools have significantly advanced financial inclusion by providing broader access to financial services.

According to the World Bank's Global Financial Inclusion, the number of adults with an account in a formal financial institution increased from 35% in 2011 to 77% in 2021 in India.

Evolution of India's Approach:

- **Strategic Shift:** India's financial inclusion strategy has evolved from targeting 'every household' to focusing on 'every adult', aiming for broader and more effective reach.
- **Enhanced Account Usage:**
 - **Direct Benefit Transfer (DBT):** Emphasis has been placed on increasing the usage of accounts through DBT flows.
 - **Digital Payment Promotion:** Promotion of digital payments via RuPay cards, UPI, and other tools to improve financial accessibility and efficiency.
- **International Collaboration:** India's participation in Project Nexus aims to enhance cross-border retail payments by interlinking domestic Fast Payment Systems (FPSs) with those of ASEAN countries.
- **Digital Financial Inclusion (DFI):** Research by the IMF indicates that DFI lead to a 2.2% increase in average economic growth, driven by higher consumption and formalization of the economy.
- **Government Initiatives:** Initiatives such as Digital India Mission, Digital Public Infrastructure (DPI), UPI, DBT, etc advanced financial inclusion.

Progress on financial inclusion in India



Key Achievements:

- **Increased Accounts:** The percentage of adults with formal financial accounts rose from 35% in 2011 to 77% in 2021.
 - **Reduced Access Gap:** There has been a reduction in the disparity in financial access between rich and poor, and narrowing the gender gap.
- **Improved Digital Financial Tools:** Tools like Aadhaar, e-KYC, UPI, and Bharat QR have been instrumental in enhancing accessibility and usability of financial services.
- **Economic Impact:** Increased DFI has contributed positively to economic growth through enhanced consumption and formal sector participation.

Challenges:

- **Digital Divide:** Despite advancements, challenges remain in ensuring that digital financial tools reach all segments of the population, particularly in remote areas.
- **Infrastructure and Security:** Ensuring robust digital infrastructure and addressing cybersecurity concerns are ongoing challenges.
- **Financial Literacy:** There is a need for improved financial literacy to enable effective use of digital financial tools among all demographic groups.
- **Regulatory and Operational Hurdles:** Navigating evolving regulatory environments and operational challenges in implementing and managing digital financial systems can be complex.

Measures to Address Challenges in Digital Financial Inclusion:

- **Expand Digital Infrastructure:** Invest in expanding digital networks and internet connectivity, particularly in remote and rural areas.
- **Enhance Cybersecurity:** Implement robust cybersecurity protocols and educate users on safe digital practices to protect against fraud and cyber threats.
- **Promote Financial Literacy:** Launch educational programs to improve financial literacy across all demographics, emphasizing the use of digital financial tools.
- **Streamline Regulations:** Simplify regulatory frameworks and provide clear guidelines to facilitate the implementation and management of digital financial systems.

Ongoing teamwork between the government, banks, and fintech companies, along with efforts to boost financial literacy and infrastructure, will be crucial for advancing digital financial inclusion in India.

Q14: Discuss the role of futures market in stabilizing prices and managing risks in Indian agriculture. Identify the challenges faced by these markets and suggest measures to address them.

(15 Marks, 250 Words)

Approach to the answer:

Introduction: Briefly explain futures markets and their function in the agricultural sector.

Body: Highlight some key initiatives and discuss how futures markets stabilize prices and help in risk management for Indian farmers, then outline challenges.

Conclusion: Suggest ways to improve futures markets.

Answer: Futures markets are financial exchanges where participants can buy and sell commodities or financial contracts for future delivery at a predetermined price. Futures markets in agriculture allow farmers and traders to lock in prices for crops before harvest, reducing the risk of price volatility and ensuring more predictable income and supply.

Key Initiatives

- **National Exchanges:** In 2003, the government established national exchanges to facilitate commodity trading, including:
 - National Commodity and Derivative Exchange (NCDEX)
 - Multi Commodity Exchange (MCX)

- National Multi Commodity Exchange (NMCE)
- **Regulatory Reforms:**
 - Commodity derivatives markets were brought under the Securities Contracts (Regulations) Act (SCRA), 1956.
 - The Securities and Exchange Board of India (SEBI) assumed regulatory control from the Forward Market Commission.
- **Electronic Online Marketplace:** The electronic National Agriculture Market (e-NAM) integrates local markets, links buyers and sellers, and ensures transparent price discovery.
- **Expansion of Eligible Commodities:** The range of commodities eligible for derivatives trading expanded from 91 to 104.

Futures markets play a significant role in stabilizing prices and managing risks:

- **Price Discovery:** Futures markets enable price discovery by reflecting market expectations, allowing farmers and traders to make informed decisions, thereby reducing price volatility.
- **Hedging Against Price Risk:** Farmers and traders can hedge against adverse price movements through futures contracts, locking in prices and securing a more predictable income.
- **Mitigating Seasonal Fluctuations:** Futures markets rule out price fluctuations caused by the imbalance between seasonal agricultural production and constant consumption, leading to more stable prices year-round and benefiting both producers and consumers.
- **Inventory Management:** It allows participants to manage inventory based on demand in the futures market, reducing the risk of overproduction or underproduction and optimizing supply chains.
- **Encouraging Investment in Agriculture:** A well-functioning futures market reduces uncertainty and risk in agricultural investments, encouraging more investment and leading to improved productivity and infrastructure.
- **Supporting Government Policies:** Futures markets provide data on future price trends that aid government policy formulation, enhancing the effectiveness of interventions for stabilizing food prices and ensuring food security.

Challenges

- **Farmer Participation:**
 - Most Indian farmers operate on a small scale, limiting their ability to participate effectively in these markets.
 - Standardized contracts with specific quality and delivery requirements are often incompatible with the diverse varieties and qualities of produce from different regions.
- **Periodic Trading Bans:** Government-imposed bans on futures trading to curb food inflation can disrupt the traded value and price volume in commodity exchanges.

Way Forward

- **Eliminate trading bans:** As markets deepen and liquidity increases, the need for banning futures trading may diminish.
- **Regulatory Review:** Regular reviews by the regulator are necessary to monitor market fluctuations and impacts.
- **Excluding Sensitive Commodities:** The 2008 Abhijit Sen Committee Report recommends identifying commodities where futures trading could influence inflation. These essential items should be excluded from futures markets until the markets are fully developed.
 - Less sensitive commodities, such as oilseed complexes (oilseeds, meals, and oils), feed (maize), cotton, basmati rice, and spices, should be the focus of the agriculture futures market.
- **Leveraging FPOs:** Farmer Producer Organizations (FPOs) can effectively link small and dispersed farmers to the commodity markets.
 - Skilling and supporting FPOs through financial literacy initiatives can encourage farmers to participate in agri-derivative markets, helping them reap the benefits of these systems.

Q15: 'Food security is not just about availability but also about accessibility and affordability.' In this context, discuss the factors affecting food security in India and suggest measures to address the situation.

(15 Marks, 250 Words)

Approach to the answer:

Introduction: Briefly define food security, emphasizing accessibility and affordability.

Body: Discuss key factors affecting food security in India, such as economic, social, and environmental challenges. Use relevant data.

Conclusion: Summarize by suggesting targeted measures to enhance accessibility and affordability in ensuring comprehensive food security.

Answer: Food security exists when all people, at all times, have physical, social and economic access to sufficient, safe, and nutritious food that meets their dietary needs and food preferences for an active and healthy life. Food security in India encompasses not just the availability of food but also its accessibility and affordability.

Data:

- In 2022, Global Food Security Index ranked India at 68/113 major countries of the world.
- As per Global Hunger Index, 2023, India's rank was 111/125 countries.

Factors affecting food security in India:

- **Economic and Production Factors:**
 - **Poverty and Unemployment:** 74% of India's population cannot afford a healthy diet (FAO, 2021), reflecting the severe impact of poverty and unemployment on food access.
 - **Low Agricultural Productivity:** Insufficient agricultural reforms and inadequate infrastructure lead to low productivity and high post-harvest losses.
 - **High Food Prices:** Inflation and inadequate food distribution systems further exacerbate the affordability crisis.
- **Poor Governance:**
 - **Corruption and Inefficiency:** Corruption in the Public Distribution System (PDS) leads to food grain diversion, denying access to those in need.
 - **Documentation Issues:** Poor state services in rural areas prevent many from accessing PDS benefits due to the lack of proper documentation.
- **Environmental and Climate Challenges:**
 - **Climate Change:** Erratic weather patterns, droughts, and heatwaves disrupt food production and threaten long-term food security.
 - **Soil Degradation:** Over-reliance on chemical inputs has led to the deterioration of soil health, with nearly 50% of cultivable land deficient in organic carbon (2023 Soil Health Survey).

Way Forward:

- **Improve PDS Efficiency:** Enhance transparency and accountability through social audits, RTI, and e-governance to curb corruption.
- **Strengthen Social Protection:** Expand initiatives like Aadhaar and ONORC to ensure food accessibility for vulnerable groups.
- **Storage and Transportation:** Develop better storage facilities including cold chain and transportation networks to reduce post-harvest losses and ensure food reaches all regions.
- **Direct Cash Transfers:** Shift from traditional input subsidies to direct cash transfers, helping farmers reduce production costs while promoting sustainable farming practices.
- **Support Small Farmers:** Consolidate landholdings through FPOs and promote contract farming to enhance productivity.
- **Buffer Stocks and Price Stabilization:** Maintain adequate buffer stocks to control price fluctuations and prevent food inflation during periods of scarcity.
- **Develop Climate-Resilient Crops:** Invest in research to create drought and flood-resistant varieties to adapt to changing environmental conditions.

- **Promote Healthy Diets:** Engage civil society and influencers to shift demand towards nutritious and sustainable food options.

By ensuring both affordability and accessibility, India can move towards a more secure and nutritionally adequate future for all.

Q16: Highlight the application of robots and automation in the agriculture sector. Identify the challenges associated with integrating these technologies into agriculture and suggest measures to address them.

(15 Marks, 250 Words)

Approach to the answer:

Introduction: Briefly introduce with some relevant data or the role of technological advancements in enhancing agricultural productivity.

Body: Identify challenges in integrating technology into agriculture and suggest measures to address these challenges.

Conclusion: Summarize the importance of overcoming these challenges for agricultural growth.

Answer: According to Mckinsey, by 2030 technological advancements could boost agricultural productivity and GDP contribution by \$600 billion, enhancing farmer incomes by 25-35% and adding \$95 billion to the economy. Embracing agtech promises a more profitable and sustainable future for Indian agriculture.

Application of Robots and Automation in the Agriculture Sector

- **Precision Seeding and Planting:** Robots equipped with GPS and sensors precisely plant seeds at optimal depths and spacing, maximizing yield and minimizing waste.
 - For Examples: Robot tractors determine the best planting locations, harvest schedules, and efficient routes for navigating farmland.
- **Automated Harvesting and Picking:** Robots with advanced vision systems identify and harvest ripe fruits and vegetables, reducing labor costs and post-harvest losses.
 - For Examples: Strawberry harvesting robots use computer vision and robotic arms to carefully pick ripe strawberries, reducing labor costs by up to 90%.
- **Intelligent Weed Control and Crop Monitoring:** Robots equipped with cameras and AI algorithms navigate fields to identify weeds and apply targeted herbicides, minimizing chemical use and protecting crops.
 - For Examples: The Agribot, developed by students from BIT Hyderabad, India, performs tasks like harvesting, spraying, seeding, and weed removal.
- **Automated Irrigation and Fertilization:** Robots equipped with soil moisture sensors apply water and fertilizer precisely where and when needed, optimizing resource use and reducing environmental impact.
- **Livestock Management and Robotics:** Robots perform tasks like automated milking, feed distribution, and animal health monitoring, enhancing efficiency and animal welfare.
 - For Examples: Automatic milking systems improve milk quality and reduce labor costs by up to 50%.

Challenges in the Application of Robots and Automation in the Agriculture Sector

- **High Initial Investment:** The high cost of these systems is often beyond the reach of small and medium-scale farmers, creating a barrier to widespread adoption.
- **Lack of Infrastructure and Connectivity:** Many rural areas lack the internet connectivity and infrastructure needed to operate and manage robots effectively, leading to data transmission issues and operational disruptions.
- **Technical Expertise and Training:** The use of complex robots and automation systems requires specialized skills and training, which are often lacking in rural areas, hindering implementation and maintenance.

- **Adaptability and Terrain Constraints:** Agricultural environments are diverse and dynamic. Robots designed for flat fields may struggle on uneven terrain or in unpredictable weather conditions, limiting their practical applications.
- **Ethical and Societal Concerns:** The potential displacement of human labor by robots raises ethical issues and social concerns. Addressing these concerns and ensuring a smooth transition for affected communities is crucial for long-term acceptance.

Way Forward

- **Fostering Innovation and Affordability:** Support innovation and reduce costs through government subsidies and grants, open-source platforms, and knowledge sharing. Utilize modular robots with interchangeable parts to enhance flexibility and reduce expenses.
- **Bridging the Infrastructure Gap:** Improve rural connectivity with initiatives like BharatNET. Implement offline data processing in robots with onboard sensors and local storage. Utilize cloud-based support and remote monitoring to ensure effective technology integration.
- **Building a Skilled Workforce:** Develop mobile training units and online courses for remote areas. Encourage peer-to-peer learning and establish mentorship programs to create a supportive network for skill development.
- **Enhancing Adaptability and Resilience:** Design robots with sensors and AI to handle various terrains, weather conditions, and crops, increasing their adaptability and acceptance in diverse agricultural environments.
- **Addressing Ethical Concerns and Ensuring Transition:** Promote the concept of robots as collaborative partners rather than replacements to address ethical concerns. For example, use robots to assist in harvesting rather than fully replace human labor.

Implementing supportive policies and enhancing farmer education will ensure successful technology integration. Initiatives like Artificial Intelligence for Agriculture Innovation (AI4AI) by the World Economic Forum is the step in right direction.

Additional Information:

Scope of Technology in the Agriculture Sector

- **Precision Agriculture:** This involves farming with high accuracy, utilizing data and technology to optimize crop yields and resource usage. *Examples:* GPS-guided tractors, drone-based monitoring.
- **Automation and Robotics:** Robots and automation are revolutionizing farming by performing labor-intensive tasks, from planting seeds to harvesting crops. *Examples:* Harvesting robots.
- **Internet of Things (IoT):** IoT devices in agriculture collect data for informed decision-making, resource optimization, and increased productivity through smart, connected technologies. *Examples:* GreenSeeker sensors, smart irrigation systems.
- **Big Data and AI:** Big data and artificial intelligence analyze extensive datasets to improve crop management, resource efficiency, and overall agricultural productivity. *Examples:* Predictive analytics, personalized farm management.
- **Gene Editing and Biotechnology:** Gene editing technology allows scientists to modify crops for desirable traits such as higher yields, pest resistance, and drought tolerance. *Examples:* CRISPR-Cas9, biofertilizers, and biopesticides.

Q17: How do orbital crowding and space debris affect space sustainability? Suggest measures to enhance sustainability of outer space. (15 Marks, 250 Words)

Approach to the answer:

Introduction: Briefly explain space sustainability and its importance.

Body: Discuss the impact of orbital crowding and space debris on sustainability. Also, suggest strategies to mitigate these issues.

Conclusion: Summarize the key measures for sustainability.

Answer: With the increasing number of satellites and other objects in orbit, orbital crowding and space debris are becoming major challenges to space sustainability. Over 26,000 objects larger than 10 cm and more than 500,000 pieces smaller than 1 cm are currently in orbit, posing significant risks to space assets.

Space Sustainability:

- Space sustainability involves the responsible and long-term use of outer space resources while preserving the space environment for current and future generations.
- It includes safe spacecraft operation, effective management of space debris and pollution, and international cooperation in space activities.

Impact of Orbital Crowding and Space Debris:

- **Increased Debris:** Orbital crowding can lead to more debris, which may damage or interfere with satellites and other spacecraft.
- **Congestion:** Crowding can make it difficult for spacecraft to maneuver and access their designated orbits.
- **Risk to Astronauts:** Increased debris can elevate radiation exposure, potentially harming astronauts and spacecraft.
- **Kessler Syndrome:** Crowded orbits can trigger Kessler Syndrome, a chain reaction where debris collisions generate more debris, further complicating space operations.

Measures for Space Sustainability:

- **International Agreement:** Establish global agreements on space resource use to ensure uniform guidelines and sustainable development.
- **Debris Minimization:** Collaborate on effective debris removal systems, such as the Space Debris Removal System (SDRS).
- **Responsible Resource Use:** Develop efficient, eco-friendly technologies and ensure resources are used safely and sustainably.
- **Reusable Technology:** Implement reusable technologies, like rockets and satellites, to reduce waste and resource consumption.

Promoting space sustainability is vital for the continued peaceful and socio-economic benefits of outer space. Implementing guidelines, ratings, and initiatives for space situational awareness and debris monitoring can help mitigate the challenges posed by orbital crowding and space debris.

Additional Information:

Additional Data:

- The **Inter-Agency Space Debris Coordination Committee (IADC)**, established in 1993, coordinates efforts among space agencies to mitigate space debris growth.
- The **European Space Agency (ESA)** has launched the **Space Safety Program** to address space debris and ensure safe, sustainable space operations.
- **ESA's Clean Space initiative** focuses on reducing the environmental impact of space missions and promoting sustainable practices within the space industry.
- The **Space Traffic Management (STM) concept** is being developed to regulate space traffic and reduce collision risks.

Indian Initiatives:

- **Project NETRA:** Initiated by ISRO, this project aims to provide real-time information on space debris status.
- **IIIT Delhi Research:** Focuses on Space Situational Awareness (SSA).
- **Digantara:** An Indian start-up planning to establish the country's first commercial SSA observatory in Uttarakhand.
- **Bilateral SSA Arrangement:** India and the US have signed an agreement to enhance SSA cooperation.
- **SPADEX:** ISRO is developing this space docking experiment for in-orbit servicing.

Q18: How does the Forest (Conservation) Amendment Act, 2023 balance economic development, border security, and forest conservation? (15 Marks, 250 Words)

Approach to the answer:

Introduction: Briefly introduce the Forest (Conservation) Amendment Act, 2023.

Body: Discuss how the Act facilitates economic development and enhances border security while ensuring forest conservation.

Conclusion: Summarize the importance of balancing development with environmental protection.

Answer: Forest (Conservation) Amendment Act of 2023 is an update of the Forest Conservation Act of 1980. The need for this amendment arises from ongoing issues such as deforestation, biodiversity loss, encroachment, forest fires, and wildlife poaching, which threaten forest health.

Benefits of the Forest (Conservation) Amendment Act, 2023:

- **Economic Development:**
 - **Provisions for Development:** The Act allows certain forest areas to be used for linear projects and security infrastructure, promoting economic growth.
 - **Empowering Local Communities:** The Act supports establishing government-owned zoos, safaris, and ecotourism outside Protected Areas. This creates job opportunities for local populations and enhances public awareness of forest and wildlife conservation.
- **Border Security:**
 - **National Security:** Projects within 100 kilometers of national borders are exempt from forest removal approvals, facilitating infrastructure development in border areas.
 - **Defense and Public Utility Projects:** Legal support is provided for land used for defense projects, paramilitary camps, or public utility projects up to five hectares in left-wing extremism-affected areas.
- **Forest Conservation:**
 - **Clarity on Forest Definition:** The Amendment clarifies that the Forest Conservation Act applies only to notified and recorded forests.
 - **Compensatory Afforestation:** The Act encourages compensatory afforestation and allows for private afforestation.

Concerns with the Forest (Conservation) Amendment Act, 2023:

- **Issues in Compensatory Reforestation:** Ambiguities in tree type specifications could prioritize financial gain over ecological sustainability in afforestation projects.
- **Prior Consent:** Removing the 'prior consent' requirement reduces tribal grama sabhas' involvement in forest conservation, potentially worsening tensions with Adivasi communities.
- **Human-Animal Conflict:** The Act lacks policies to address human-animal conflicts in Adivasi areas, increasing risks.
- **Implementation of Forest Rights Act (FRA):** State governments might reduce forest areas instead of addressing issues related to the FRA, limiting Adivasi empowerment.
- **Economy and Security Prioritization:** Exemptions for certain land uses may lead to environmental concerns being secondary to economic and security interests.

Way Forward:

- **Strengthen Enforcement Mechanisms:** Enhance enforcement with stringent penalties for violations to deter illegal activities such as deforestation and unauthorized land use.
- **Community Engagement in Sustainable Forest Management:** Involve local communities in decision-making to foster ownership and responsibility in forest management.
- **Capacity Building and Training:** Invest in the training and capacity building of forest officials, law enforcement, and local communities to improve forest conservation efforts.
- **Adaptation to Environmental Challenges:** Establish a mechanism for regular reviews and updates to the Act, allowing it to adapt to changing environmental conditions and emerging threats.

The Forest Conservation Amendment Act of 2023 is a significant step in aligning India's environmental policies with global commitments. To address associated issues, the Act should incorporate stakeholder consultations, transparency in decision-making, periodic reviews, empowerment of local communities, and environmental impact assessments, making it more effective in achieving sustainable forest management.

Q19: Quantum computers offer significant advancements in computing technology but also introduce new security challenges. Evaluate India's readiness for the quantum era. (15 Marks, 250 Words)

Approach to the answer:

Introduction: Briefly introduce quantum computing.

Body: Highlight its potential advancements and how it led to New Security Threats then discuss challenges and preparedness.

Conclusion: Summarize India's position and future steps needed to embrace the quantum era.

Answer: Quantum computers utilize the principles of quantum physics to store data and perform computations. Unlike classical computers, which encode information in binary "bits" (0s or 1s), quantum computers use quantum bits or qubits as their basic unit of memory. Examples include IBM's Q System One, Google's Sycamore processor, and Honeywell's System Model H1.

Quantum - Technical Breakthroughs

- **Faster Problem Solving:** Quantum computers can solve complex mathematical problems quickly, which is crucial for fields like big data analysis.
 - For Example: In 2019, Google achieved quantum supremacy by solving a specific computational task in 200 seconds.
- **Enhanced Pattern Recognition:** Quantum computing improves pattern recognition, vital for AI applications in sectors such as healthcare and finance.
 - For Example: Japanese researchers developed a quantum neural network that detects credit card fraud with 63% accuracy compared to 50% for classical networks.
- **Breakthroughs in Drug Discovery:** Quantum computers can simulate molecular structures with high precision, aiding in the discovery of new materials and drugs.
 - For Example: SRI International and AstraZeneca are exploring quantum computing's potential in protein modeling.
- **Financial Analysis:** Quantum computing can significantly enhance portfolio optimization, price modeling, risk analysis, and other financial services.
- **Optimization of Chemical Reactions:** Quantum computing helps develop more efficient and sustainable industrial processes.
 - For Example: Volkswagen used quantum computing to optimize the chemical composition of car batteries.
- **Secure Communication:** Quantum cryptography offers theoretically unbreakable encryption.
 - For Example: China has built a quantum key distribution (QKD) network spanning thousands of kilometers.
- **Accurate Weather Predictions:** Quantum computers can process complex weather models faster.
 - For Example: NASA plans to use quantum computing for high-resolution climate modeling.

New Security Threats

- **Cyber Risks:** Quantum computers pose a significant threat to current cybersecurity protocols due to their immense computational power.
 - For Example: Cyber attackers might be storing encrypted data to decrypt it in the future using quantum computers ("harvest now, decrypt later").
- **Breaking Blockchain Security:** Quantum computing can compromise the security of blockchain technologies and cryptocurrencies.
 - For Example: A 2019 study by the University of Sussex highlighted the vulnerabilities of Bitcoin to future quantum attacks.

- **Eavesdropping on Encrypted Communications:** Quantum computers have the potential to decrypt messages intercepted from supposedly secure channels.
- **Vulnerability of Web Browsing & VPN:** The security of HTTPS and Virtual Private Networks (VPNs) could be compromised by quantum computing.
- **Forgery of Digital Signatures:** Quantum computers could forge digital signatures, making it possible to alter documents and transactions.

India's Preparedness

- **National Quantum Mission (NQM):** The mission aims to develop intermediate-scale quantum computers with 50-100 physical qubits in 5 years and 50-1000 physical qubits in 8 years.
- **Emerging Companies:** Startups like Qunu Labs in Bangalore and BosonQ in Bhilai are making significant contributions to quantum technology.
- **Public-Private Partnership:** MeitY and AWS have announced a Quantum Computing Applications Lab.
 - For Example: Samsung Semiconductor India Research and IISc are setting up a quantum technology lab.
- **Promotion of Research & Development:**
 - In 2021, the Indian Army established a quantum computing laboratory.
 - The Centre for Development of Telematics (C-DOT) launched a quantum communication lab in 2021.
 - IISc established a dedicated Quantum Technologies and Application Centre to conduct research on NISQ computing software.

Way Forward

- **Building Quantum Research Facilities:** Expand and enhance quantum research facilities across research institutions.
- **International Cooperation:** Partner with leading global institutions and countries to share knowledge.
- **Support for Startups:** Offer financial incentives, incubation, and mentorship to quantum technology startups.
- **Regulatory and Policy Framework:** Establish national and international standards for quantum technology.

India's proactive investments and strategic initiatives in quantum technology will position it to lead future advancements while fostering economic growth and global innovation.

Q20: Crypto currencies can pose serious internal security threat to India and the world at large. Discuss the steps taken by the government to tackle the challenges. (15 Marks, 250 Words)

Approach to the answer:

Introduction: Briefly introduce the concept of cryptocurrencies and their potential security threats.

Body: Discuss the internal security challenges posed by cryptocurrencies. Outline the steps taken by the government to address these challenges.

Conclusion: Summarize the need for robust regulatory frameworks to mitigate security risks.

Answer: Crypto currency is a virtual currency that is secured by cryptography. They are decentralized networks, based on block-chain technology—a distributed ledger enforced by a disparate network of computers. As per Chainalysis, illicit transactions using crypto currencies in 2021 increased by 79% compared to 2020, making it among most popular tool with criminals for nefarious activities.

Internal security threat to India:

- **Terrorism:** There is wider use of crypto currencies on the dark net (like TOR, Freenet, Zeronet) for terror acts and drug trafficking.

- Money Laundering: Anonymity and lack of regulation makes it preferred tool to launder money across borders. Recent Chainalysis report noted that Money laundering in Ethereum was 63 per cent, and 57 per cent in Stable coins.
- Extortion: -Crypto currency is sometimes used as payment for extortion because the money trail is highly encrypted and cannot be traced easily.
- Tax evasion: -Anonymity offered by the crypto currency ecosystem makes it difficult for tax authorities to trace transactions to individuals and verify their tax liabilities.

Security threat to the World:

- Use of cryptocurrencies to finance terror acts, procure weapons and explosives without being tracked by intelligence agencies.
- Criminals can infect individual computers with malwares to steal crypto currency and personal information.
- High fluctuation in value of crypto currencies is a threat to economies across the globe causing instability.
- Small economies like El Salvador and Central African Republic have made Bitcoin as legal tender, further increasing their economic vulnerability.
- Environment: - Crypto miners employ sophisticated, energy intensive machines and use sometimes as much energy as entire countries consume. Energy used has a large portion come from coal-fired power plants.

Currently, there are no national guidelines on crypto currency related cases, due to which enforcement agencies often struggle, particularly in seizure as well as tracing suspects. Thus, government took certain steps to tackle this menace.

Steps taken to tackle the challenges:

- Initially, in 2016, the government outright banned cryptocurrencies.
- Presently the government and parliamentary committee are engaging with experts and other stakeholders to frame regulations around crypto currencies.
- The RBI had expressed its opinion to ban cryptocurrencies.
- The Union Budget 2022-2023 of India has also proposed to introduce a digital currency in the coming financial year.
- Finance ministry has clearly said that cryptocurrencies like Bitcoin or Ethereum will never become legal tender.
- In Budget 2022, it was announced that “any income from transfer of any virtual digital asset shall be taxed at the rate of 30%.

Suggestions:

- **Crypto wallet for agencies:** As per Bureau of Police Research and Development guidelines, law enforcement agencies must have their own crypto wallet to store seized virtual digital assets.
- **Fraud detection:** Block-chain technologies should implement automated Anti Money Laundering (AML) fraud detection that would block or flag suspicious transactions prior to them being carried out.
- **Awareness:** There is need to educate and inform people to reduce the risk associated with crypto currency
- Globally, nations need to collaborate to tackle the challenges due to crypto currencies. UNODC has launched training to tackle money laundering and bitcoin banking fraud.

A comprehensive Crypto currency Bill would as a first step go a long way in establishing a sound regulatory and inclusive regime for the crypto currency ecosystem and facilitate international cooperation to ensure that competent authorities are able to act rapidly and effectively.